

Assignment 5: Data Visualization

Laura Piccirillo

Fall 2025

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file <FirstLast>_A05_DataVisualization.Rmd (replacing <FirstLast> with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the NEON_NIWO_Litter_mass_trap_Processed.csv version, again from the Processed_KEY folder).
2. Make sure R is reading dates as date format; if not change the format to date.

```
#1 installing and loading packages, checking working directory
#install.packages("tidyverse")
#install.packages("here")
#install.packages("cowplot")
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2    3.5.1      v tibble     3.2.1
## v lubridate  1.9.4      v tidyr      1.3.1
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag() masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## here() starts at /home/guest/ENV 872
```

```
library(cowplot)
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
## stamp
```

```
getwd()
```

```
## [1] "/home/guest/ENV 872"
```

```
here()
```

```
## [1] "/home/guest/ENV 872"
```

```
PeterPaul <- read.csv(
  here("Data/Processed_KEY/NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv"),
  stringsAsFactors = TRUE)
```

```
litter <- read.csv(
  here("Data/Processed_KEY/NEON_NIWO_Litter_mass_trap_Processed.csv"),
  stringsAsFactors = TRUE)
```

```
#2 checking date classes for datasets
class(PeterPaul$sampdate)
```

```
## [1] "factor"
```

```
class(litter$collectDate)
```

```
## [1] "factor"
```

```
PeterPaul$sampdate <- ymd(PeterPaul$sampdate)
litter$collectDate <- ymd(litter$collectDate)
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```
#3

plot_theme <-
  theme_bw() + theme(axis.text = element_text(color = "black"),
                     legend.position = "top")
```

Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

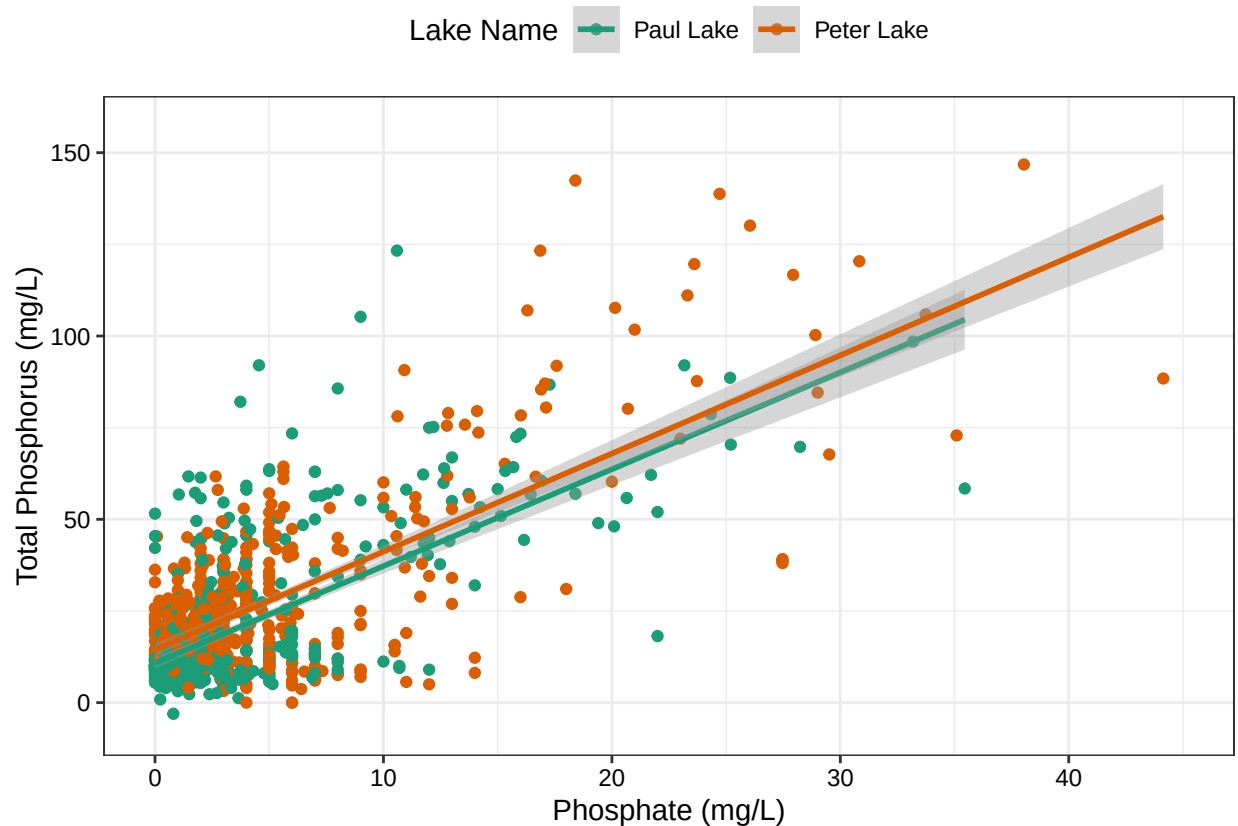
4. [NTL-LTER] Plot total phosphorus (tp_ug) by phosphate (po4), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

```
#4 lake phosphorus plotting
ggplot(PeterPaul, aes(po4, tp_ug, color = lakename)) + geom_point() +
  geom_smooth(method = lm) + xlim(0, 45) + plot_theme +
  xlab("Phosphate (mg/L)") + ylab("Total Phosphorus (mg/L)") +
  labs(color = "Lake Name") + scale_color_brewer(palette = "Dark2")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 21947 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

```
## Warning: Removed 21947 rows containing missing values or values outside the scale range
## ('geom_point()').
```



5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

```
#5

temp_plot <- ggplot(PeterPaul, aes(x=factor(month, levels=1:12,
                                           labels=month.abb),
                                   y = temperature_C,
                                   colour = lakename)) + geom_boxplot() +
  scale_x_discrete(name='Month', drop=FALSE) + plot_theme +
  ylab("Temperature (celsius)") + labs(color = "Lake Name") +
  scale_color_brewer(palette = "Dark2") + theme(axis.title.x = element_blank())

TP_plot <- ggplot(PeterPaul, aes(x=factor(month, levels=1:12,
                                           labels=month.abb),
                                   y = tp_ug,
                                   colour = lakename)) + geom_boxplot() +
  scale_x_discrete(name='Month', drop=FALSE) + plot_theme +
  ylab("Total Phosphorus (mg/L)") + labs(color = "Lake Name") +
```

```

scale_color_brewer(palette = "Dark2") + theme(axis.title.x = element_blank(),
                                              legend.position = "none")

TN_plot <- ggplot(PeterPaul, aes(x=factor(month, levels=1:12,
                                          labels=month.abb),
                                y = tn_ug,
                                colour = lakename)) + geom_boxplot() +
  scale_x_discrete(name='Month', drop=FALSE) + plot_theme +
  ylab("Total Nitrogen (mg/L)") + labs(color = "Lake Name") +
  scale_color_brewer(palette = "Dark2") + theme(legend.position = "none")

plot_grid(temp_plot, TP_plot, TN_plot, ncol = 1, align = "v")

```

```

## Warning: Removed 3566 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

```

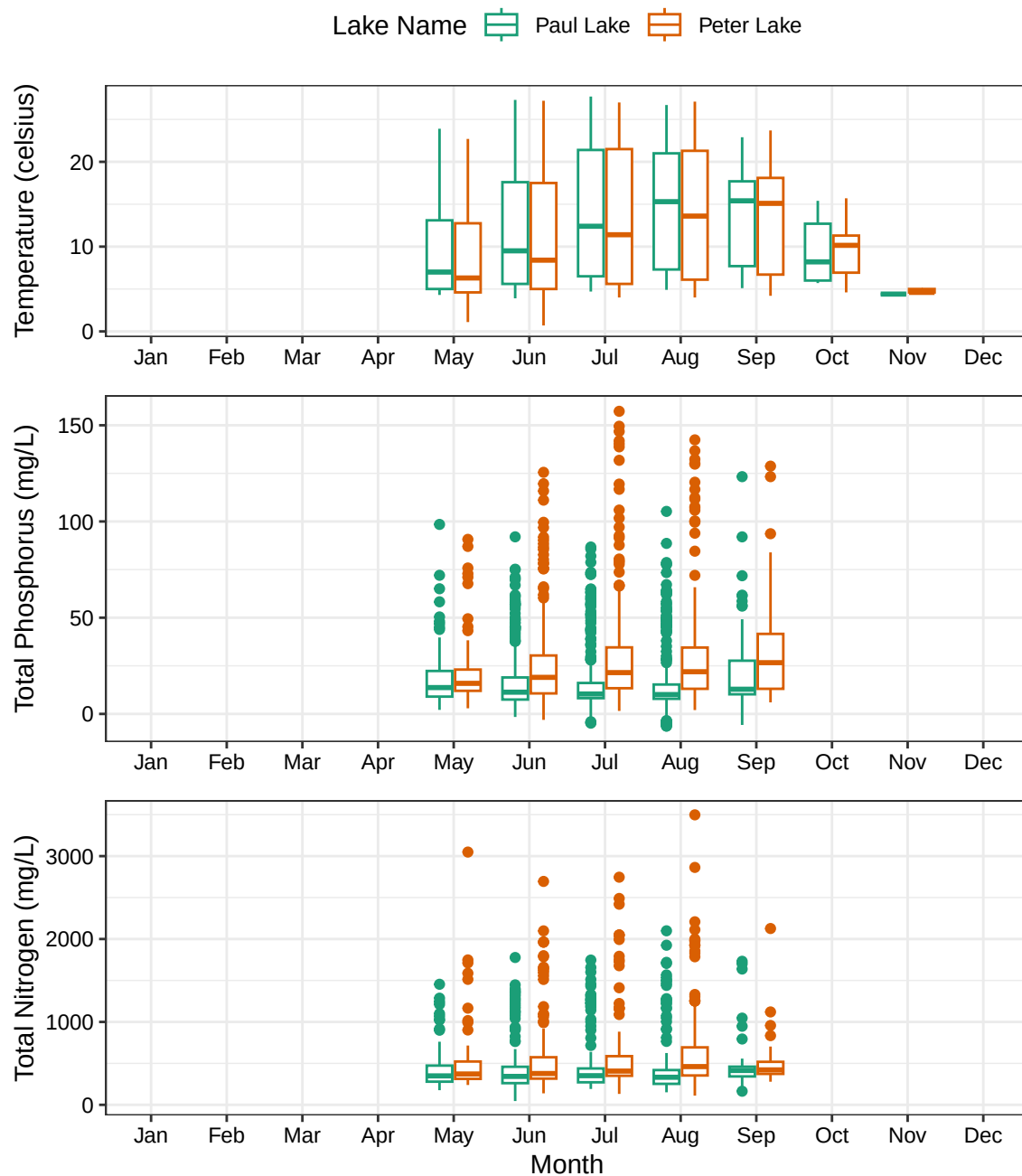
## Warning: Removed 20729 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

```

## Warning: Removed 21583 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```



Question: What do you observe about the variables of interest over seasons and between lakes?

Answer:

6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the “Needles” functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

#6

#7

Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

Answer: